IEEE P802.1CBec/D0.2
(Draft amendment to IEEE Std 802.1CB™-2017, as amended by
IEEE P802.1CBcv-2021, and IEEE P802.1CBbd-2021)
October 23, 2025

Draft Standard for Local and metropolitan area networks — Frame Replication and Elimination for Reliability — Amendment:

Guidance for Sequence Recovery Function Parameter Configuration

Prepared by the Time-Sensitive Networking Task Group of IEEE 802.1

LAN MAN Standards Committee of the **IEEE Computer Society**

Copyright © 2025 by the Institute of Electrical and Electronics Engineers, Inc.
Three Park Avenue
New York, New York 10016-5997, USA

All rights reserved.

Important Notice

This document is an unapproved draft of a proposed IEEE Standard. IEEE hereby grants the named IEEE SA Working Group or Standards Committee Chair permission to distribute this document to participants in the receiving IEEE SA Working Group or Standards Committee, for purposes of review for IEEE standardization activities. No further use, reproduction, or distribution of this document is permitted without the express written permission of IEEE Standards Association (IEEE SA). Prior to any review or use of this draft standard, in part or in whole, by another standards development organization, permission must first be obtained from IEEE SA (stds-copyright@ieee.org). This page is included as the cover of this draft, and shall not be modified or deleted.

IEEE Standards Association
445 Hoes Lane
Piscataway, NJ 08854, USA

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Abstract: This amendment adds an informative annex describing recommended values for the existing sequence recovery function parameters, and provides guidance regarding frame buffering in relay and end systems to assist in the usage of Frame Replication and Elimination for Reliability. This amendment also includes technical and editorial corrections in the description of existing IEEE Std 802.1CB functionality.

Keywords: TSN, Time-Sensitive Networking, Redundancy, Bridging, Bridges, Frame Replication, Frame Elimination, Sequence Recovery Function, Bridged Local Area Networks, IEEE 802®, IEEE 802.1Q™, IEEE 802.1CB™, local area networks (LANs), MAC Bridges, Virtual Bridged Local Area Networks (virtual LANs), amendment.

Copyright © 2022 IEEE. All rights reserved.
This cover page is not part of the draft standard

Important Notices and Disclaimers Concerning IEEE Standards Documents

IEEE documents are made available for use subject to important notices and legal disclaimers. These notices and disclaimers, or a reference to this page, appear in all standards and may be found under the heading "Important Notice" or "Important Notices and Disclaimers Concerning IEEE Standards Documents."

Notice and Disclaimer of Liability Concerning the Use of IEEE Standards Documents

IEEE Standards documents (standards, recommended practices, and guides), both full-use and trial-use, are developed within IEEE Societies and the Standards Coordinating Committees of the IEEE Standards Association ("IEEE-SA") Standards Board. IEEE ("the Institute") develops its standards through a consensus development process, approved by the American National Standards Institute ("ANSI"), which brings together volunteers representing varied viewpoints and interests to achieve the final product. Volunteers are not necessarily members of the Institute and participate without compensation from IEEE. While IEEE administers the process and establishes rules to promote fairness in the consensus development process, IEEE does not independently evaluate, test, or verify the accuracy of any of the information or the soundness of any judgments contained in its standards.

IEEE does not warrant or represent the accuracy or content of the material contained in its standards, and expressly disclaims all warranties (express, implied and statutory) not included in this or any other document relating to the standard, including, but not limited to, the warranties of: merchantability; fitness for a particular purpose; non-infringement; and quality, accuracy, effectiveness, currency, or completeness of material. In addition, IEEE disclaims any and all conditions relating to: results; and workmanlike effort. IEEE standards documents are supplied "AS IS" and "WITH ALL FAULTS."

Use of an IEEE standard is wholly voluntary. The existence of an IEEE standard does not imply that there are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and issued is subject to change brought about through developments in the state of the art and comments received from users of the standard.

In publishing and making its standards available, IEEE is not suggesting or rendering professional or other services for, or on behalf of, any person or entity nor is IEEE undertaking to perform any duty owed by any other person or entity to another. Any person utilizing any IEEE Standards document, should rely upon his or her own independent judgment in the exercise of reasonable care in any given circumstances or, as appropriate, seek the advice of a competent professional in determining the appropriateness of a given IEEE standard.

IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO: PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE UPON ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Translations

The IEEE consensus development process involves the review of documents in English only. In the event that an IEEE standard is translated, only the English version published by IEEE should be considered the approved IEEE standard.

Official statements

A statement, written or oral, that is not processed in accordance with the IEEE-SA Standards Board Operations Manual shall not be considered or inferred to be the official position of IEEE or any of its committees and shall not be considered to be, or be relied upon as, a formal position of IEEE. At lectures, symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall make it clear that his or her views should be considered the personal views of that individual rather than the formal position of IEEE.

Comments on standards

Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of membership affiliation with IEEE. However, IEEE does not provide consulting information or advice pertaining to IEEE Standards documents. Suggestions for changes in documents should be in the form of a proposed change of text, together with appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is important that any responses to comments and questions also receive the concurrence of a balance of interests. For this reason, IEEE and the members of its societies and Standards Coordinating Committees are not able to provide an instant response to comments or questions except in those cases where the matter has previously been addressed. For the same reason, IEEE does not respond to interpretation requests. Any person who would like to participate in revisions to an IEEE standard is welcome to join the relevant IEEE working group.

Comments on standards should be submitted to the following address:

Secretary, IEEE-SA Standards Board

445 Hoes Lane

Piscataway, NJ 08854 USA

Laws and regulations

Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the provisions of any IEEE Standards document does not imply compliance to any applicable regulatory requirements. Implementers of the standard are responsible for observing or referring to the applicable regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not in compliance with applicable laws, and these documents may not be construed as doing so.

Copyrights

IEEE draft and approved standards are copyrighted by IEEE under U.S. and international copyright laws. They are made available by IEEE and are adopted for a wide variety of both public and private uses. These include both use, by reference, in laws and regulations, and use in private self-regulation, standardization, and the promotion of engineering practices and methods. By making these documents available for use and adoption by public authorities and private users, IEEE does not waive any rights in copyright to the documents.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Photocopies

Subject to payment of the appropriate fee, IEEE will grant users a limited, non-exclusive license to photocopy portions of any individual standard for company or organizational internal use or individual, non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400. Permission to photocopy portions of any individual standard for educational classroom use can also be obtained through the Copyright Clearance Center.

Updating of IEEE Standards documents

Users of IEEE Standards documents should be aware that these documents may be superseded at any time by the issuance of new editions or may be amended from time to time through the issuance of amendments, corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the document together with any amendments, corrigenda, or errata then in effect.

Every IEEE standard is subjected to review at least every ten years. When a document is more than ten years old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that they have the latest edition of any IEEE standard.

In order to determine whether a given document is the current edition and whether it has been amended through the issuance of amendments, corrigenda, or errata, visit the IEEE-SA Website at https://ieeexplore.ieee.org/browse/standards/collection/ieee or contact IEEE at the address listed previously. For more information about the IEEE SA or IEEE's standards development process, visit the IEEE-SA Website at http://standards.ieee.org.

Errata

Errata, if any, for all IEEE standards can be accessed on the IEEE-SA Website at the following URL: https://standards.ieee.org/standard/index.html. Users are encouraged to check this URL for errata periodically.

Patents

Attention is called to the possibility that implementation of this standard may require use of subject matter covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the IEEE-SA Website at http://standards.ieee.org/about/sasb/patcom/patents.html. Letters of Assurance may indicate whether the Submitter is willing or unwilling to grant licenses under patent rights without compensation or under reasonable rates, with reasonable terms and conditions that are demonstrably free of any unfair discrimination to applicants desiring to obtain such licenses.

Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not responsible for identifying Essential Patent Claims for which a license may be required, for conducting inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their own responsibility. Further information may be obtained from the IEEE Standards Association.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Participants

The following is a list of participants in the IEEE 802.1 Working Group.

Glenn Parsons, *Chair*
Jessy Rouyer, *Vice-Chair*
János Farkas, *Chair, Time-Sensitive Networking Task Group*
Lisa Maile, *Editor*

The following members of the [individual/entity] balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

When the IEEE-SA Standards Board approved this standard on XX Month 20XX, it had the following membership:

Name, *Chair*
Name, *Vice Chair*
Name, *Past Chair*

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Name, *Secretary*

SBMember1	SBMember9	SBMember17
SBMember2	SBMember10	SBMember18
SBMember3	SBMember11	SBMember19
SBMember4	SBMember12	SBMember20
SBMember5	SBMember13	SBMember21
SBMember6	SBMember14	SBMember22
SBMember7	SBMember15	SBMember23
SBMember8	SBMember16	

*Member Emeritus

Also included are the following nonvoting IEEE-SA Standards Board liaisons:

Name, *DOE Representative*
Name, *NIST Representative*

Name
IEEE Standards Program Manager, Document Development

Name
IEEE Standards Program Manager, Technical Program Development

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Introduction to IEEE P802.1CBec

This introduction is not part of IEEE P802.1CBec/D0.2, Draft Standards for Local and Metropolitan Area Networks—Amendment: Guidance for Sequence Recovery Function Parameter Configuration

This Standard defines Guidance for Sequence Recovery Function Parameter Configuration.

This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution. Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and to incorporate new related material. Information on the current revision state of this and other IEEE 802 standards can be obtained from

Secretary, IEEE-SA Standards Board

445 Hoes Lane

Piscataway, NJ 08854, USA

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Editor's Foreword

<<Notes>>

<<Throughout this document, all notes such as this one, presented between angle braces, are temporary notes inserted by the Editors for a variety of purposes; these notes and the Editors' Foreword will all be removed prior to publication and are not part of the normative text.>>

<<Comments and participation in 802.1 standards development

Comments on this draft are encouraged! ***PLEASE NOTE: All issues related to IEEE standards presentation style, formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to publication, after balloting and the process of achieving agreement on the technical content of the standard is complete.*** *Readers are urged to devote their valuable time and energy only to comments that materially affect either the technical content of the document or the clarity of that technical content. Comments should not simply state what is wrong, but also what might be done to fix the problem.*

Full participation in the development of this draft requires individual attendance at IEEE 802 meetings. Information on 802.1 activities, working papers, and email distribution lists etc. can be found on the 802.1 website:

http://ieee802.org/1/

Use of the email distribution list is not presently restricted to 802.1 members, and the working group has had a policy of considering ballot comments from all who are interested and willing to contribute to the development of the draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity in past projects. Non-members are advised that the email lists exist primarily to allow the members of the working group to develop standards, and are not a general forum.

All participants in IEEE standards development have responsibilities under the IEEE patent policy and should familiarize themselves with that policy. See http://standards.ieee.org/about/sasb/patcom/materials.html.

Comments on this document may be sent to the 802.1 email exploder, to the editors, or to the Chairs of the 802.1 Working Group and Time-Sensitive Networking Task Group.

Lisa Maile
Editor
Email: lisa.maile@ieee.org

János Farkas	*Glenn Parsons*
Chair, 802.1 Time-Sensitive Networking Task Group	*Chair, 802.1 Working Group*
Email: janos.farkas@ericsson.com	*Email: glenn.parsons@ericsson.com*

PLEASE NOTE: Comments whose distribution is restricted in any way cannot be considered, and may not be acknowledged. >>

<< Introductory notes to P802.1CBec Draft 0.2

Draft 0.2 is prepared for the second Task Group ballot.The major changes from D0.1 are as follows:

- *Terminology clarified (e.g., "first received packet", best-case delay, v_S)*
- *Individual Recovery function explicitly excluded*

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

— *Consequences clearer explained*
— *VectorRecoveryAlgorithm behavior illustrated*
— *Text restructured for clarity* >>

<< Introductory notes to P802.1CBec Draft 0.1

Draft 0.1 was prepared by Lisa Maile for the initial Task Group ballot. The text contribution has been reviewed at the TSN TG meeting held on 2025-04-24. Based on the comments received during this presentation, updates have been included in this draft. >>

<< Project Authorization Request, Scope, Purpose, and Five Criteria

A PAR (Project Authorization Request) for P802.1CBec was approved by the IEEE Standards Association on February 13, 2025. The following information is taken from the 802.1CBec PAR and Five Criteria.

Scope of Proposed Project:

This amendment adds an informative annex describing recommended values for the existing sequence recovery function parameters, and provides guidance regarding frame buffering in relay and end systems to assist in the usage of Frame Replication and Elimination for Reliability. This amendment also includes technical and editorial corrections in the description of existing IEEE Std 802.1CB functionality.

Purpose of Proposed Project:

This document will not include a purpose clause.

Need for the Proposed Project:

The sequence recovery function specified in the base standard eliminates duplicate frames and forwards non-duplicate frames. Inappropriate setting of parameter values and buffer dimensioning could lead to duplicates being passed and unintentional frame elimination. This informative annex provides guidance to avoid this unwanted behavior.

The remaining information is taken from the Criteria for Standards Development (CSD).

1. IEEE 802 criteria for standards development (CSD)

The CSD documents an agreement between the WG and the Sponsor that provides a description of the project and the Sponsor's requirements more detailed than required in the PAR. The CSD consists of the project process requirements, 1.1, and the 5C requirements, 1.2.

1.1 Project process requirements

1.1.1 Managed objects

Describe the plan for developing a definition of managed objects. The plan shall specify one of the following:

a) The definitions will be part of this project.

b) The definitions will be part of a different project and provide the plan for that project or anticipated future project.

c) The definitions will not be developed and explain why such definitions are not needed.

This project will use method c). The definitions will not be developed, since this amendment will only add an informative annex to the base standard.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

1.1.2 Coexistence

A WG proposing a wireless project shall demonstrate coexistence through the preparation of a Coexistence Assurance (CA) document unless it is not applicable.

a) Will the WG create a CA document as part of the WG balloting process as described in Clause 13? (yes/no)

b) If not, explain why the CA document is not applicable.

Item b) is applicable. This project is not a wireless project; therefore, the CA document is not applicable.

1.2 5C requirements

1.2.1 Broad market potential

Each proposed IEEE 802 LMSC standard shall have broad market potential. At a minimum, address the following areas:

a) Broad sets of applicability.

b) Multiple vendors and numerous users.

a) The proposed amendment is applicable in the various use cases where Frame Replication and Elimination for Reliability is used to improve reliability and availability.

b) All vendors and users of Frame Replication and Elimination for Reliability can benefit from the informative annex.

1.2.2 Compatibility

Each proposed IEEE 802 LMSC standard should be in conformance with IEEE Std 802, IEEE 802.1AC, and IEEE 802.1Q. If any variances in conformance emerge, they shall be thoroughly disclosed and reviewed with IEEE 802.1 WG prior to submitting a PAR to the Sponsor.

a) Will the proposed standard comply with IEEE Std 802, IEEE Std 802.1AC and IEEE Std 802.1Q?

b) If the answer to a) is no, supply the response from the IEEE 802.1 WG.

The informative annex will be in conformance with IEEE Std 802, IEEE Std 802.1AC, and IEEE Std 802.1Q.

1.2.3 Distinct Identity

Each proposed IEEE 802 LMSC standard shall provide evidence of a distinct identity. Identify standards and standards projects with similar scopes and for each one describe why the proposed project is substantially different.

No existing IEEE 802 standard or standard project addresses the recommended setting of parameters for the sequence recovery function of Frame Replication and Elimination for Reliability.

1.2.4 Technical Feasibility

Each proposed IEEE 802 LMSC standard shall provide evidence that the project is technically feasible within the time frame of the project. At a minimum, address the following items to demonstrate technical feasibility:

a) Demonstrated system feasibility.

b) Proven similar technology via testing, modeling, simulation, etc.

The informative annex provides formulas to determine a recommended set of values to use for existing parameters. All addressed parameters are already defined by IEEE Std 802.1CB; therefore, setting these parameters is feasible.

The informative annex relies on the proven technology provided by the base standard. Additionally, the proposed parameter values have been modeled, proven, and simulated.

See https://ieeexplore.ieee.org/document/9838905 (http://arxiv.org/abs/2306.13469).

1.2.5 Economic Feasibility

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Each proposed IEEE 802 LMSC standard shall provide evidence of economic feasibility. Demonstrate, as far as can reasonably be estimated, the economic feasibility of the proposed project for its intended applications. Among the areas that may be addressed in the cost for performance analysis are the following:

a)Balanced costs (infrastructure versus attached stations).

b)Known cost factors.

c)Consideration of installation costs.

d)Consideration of operational costs (e.g., energy consumption).

e)Other areas, as appropriate.

This project will not add new functionality, hence no impact on cost or complexity. The amendment does not imply additional hardware cost as it only provides formulas to determine a recommended set of values for setting existing parameters. The provided recommended set of values to use for the existing parameters and the provided guidelines could potentially lower implementation costs by reducing over-provisioning for memory (required for the sequence recovery function and buffering).

>>

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Contents

Annex X (informative) Guidance for Sequence Recovery Function Parameter Configuration 15

X.1 Introduction 15
X.2 Terminology 15
X.3 Consequences of Parameter Settings 17
X.4 Parameter Configuration 17
X.4.1 Recovery Algorithm Selection 17
X.4.1.1 Periodic Streams 17
X.4.1.2 Interval-constrained Streams and Multiple Packets 17
X.4.2 Sequence History Length 18
X.4.2.1 Periodic Streams 18
X.4.2.2 Interval-constrained Streams 19
X.4.2.3 Multiple Packets 19
X.4.3 Reset Timer Configuration 19
X.4.3.1 Periodic Streams 19
X.4.3.2 Interval-constrained Streams 20
X.4.3.3 Multiple Packets 20

Annex Y (informative) Bibliography 21

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Draft Standard for Local and metropolitan area networks — Frame Replication and Elimination for Reliability — Amendment:

Guidance for Sequence Recovery Function Parameter Configuration

This amendment is based on IEEE Std 802.1CB™-2017

NOTE—The editing instructions contained in this amendment define how to merge the material contained here into the base document and its other amendments to form the new comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: ***change***, ***delete***, ***insert***, and ***replace***. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed either by using ~~strikethrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. **Insertion** may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.[1]

[1]Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Annex X

(informative)

Guidance for Sequence Recovery Function Parameter Configuration

<< Editor's note. The PAR and CSD for IEEE P802.1CBec are under consideration for modification (cf. TSN TG, September 2025 Interim Session). The proposed changes are expected to affect the normative definition of the Sequence Recovery function, with current discussions primarily focusing on the reset function. The guidelines in this annex will be subject to change following any resulting normative changes.>>

X.1 Introduction

Incorrect parameter settings can compromise the reliability of FRER. This annex provides informative guidance for the configuration of parameters of the Sequence recovery function (7.4.2). The content of this annex is based on concepts and results presented in [B1].

This annex covers the configuration of the following parameters:

— frerSeqRcvyAlgorithm (10.4.1.5)
— frerSeqRcvyHistoryLength (10.4.1.6)
— frerSeqRcvyResetMSec (10.4.1.7)

An overview of potential consequences that can arise from incorrect parameter settings is provided in Section X.3; details on suggested configuration settings are discussed in Section X.4.

X.2 Terminology

The following variables are used throughout this annex:

a) T_S : Minimum duration of the transmission interval at the talker associated with the Stream. This refers to the duration over which a Stream may transmit up to N_S packets. This variable can correspond to the Class Measurement Interval (see IEEE Std 802.1Q™-2022, Clause L.2). It is defined per stream, not per class.
b) N_S : Maximum number of packets a Stream is allowed to transmit during one transmission interval T_S. This variable can correspond to Max Interval Frames (see IEEE Std 802.1Q™-2022, Clause 46.2.3.5.2).
c) V_S : Maximum deviation in the packet generation time at the talker, e.g., due to clock inaccuracy. The deviation is assumed to be bounded and non-accumulating. If the transmission of a packet can occur up to V_1 later or up to V_2 earlier than its nominal time, the maximum deviation is defined as $V_S = V_1 + V_2$.

This annex uses the following terminology for the description of Sequence Recovery behavior: *First received packet of a given Sequence Number* is the first packet of a Sequence Number that is received at the Sequence recovery function. This packet should be forwarded. Any subsequent packet with the same Sequence Number is named *replicate*. Replicates should be discarded.

The guideline distinguishes between two classes of Streams:

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

d) *Periodic Streams* that generate N_S packets in a strictly periodic pattern. The pattern repeats every T_S time units, with a maximum deviation V_S from the expected time.

e) *Interval-constrained Streams* that generate up to N_S packets during each interval of length T_S. The intervals are accurate up to $\pm V_S$ per stream. Only the number of packets and the interval duration per stream are constrained; no assumptions are made on the inter-packet spacing.

The two classes are illustrated in Figure X-1 for $N_S = 3$ and $V_S = 0$.

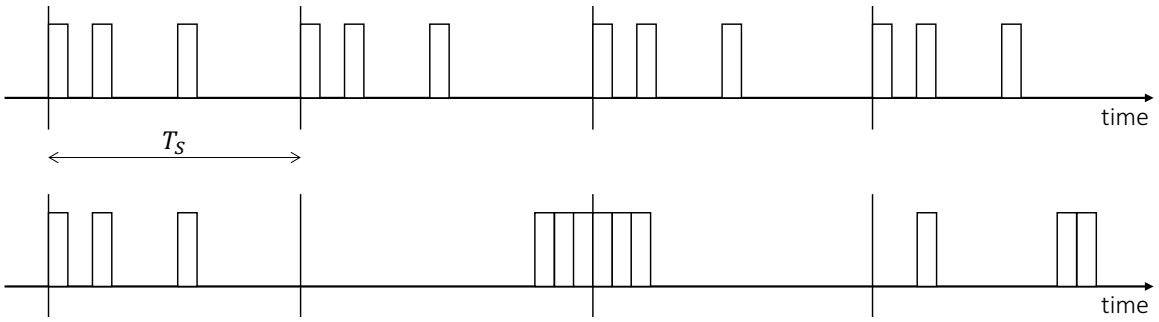


Figure X-1—Example for Talker sending behavior.
Top: Periodic Stream. Bottom: Interval-constrained Stream.

The following delay values are assumed to be known. Their determination is out of the scope of this annex:

f) d_{BC} : Best-case delay, the minimum time from the sending of a packet to its earliest arrival at the Sequence recovery function, considering any packet of any Member Stream.

g) d_{WC}: Worst-case delay, the maximum time from the sending of a packet to the arrival of its last copy at the Sequence recovery function, considering any packet of any Member Stream.

h) $\Delta d = d_{WC} - d_{BC}$: Reception interval, represents the time interval during which packets with identical Sequence Numbers may arrive at the Sequence recovery function, considering any packet of any Member Stream.

Figure X-2 illustrates the concept of reception interval. The following guidelines base on modeling the worst-case overlap of reception intervals for every parameter.

For simplicity, all variables are defined from the Talker to the Sequence Recovery function. Alternatively, items a)–c) and f)–h) may also be defined from the replication to the Sequence Recovery function for increased precision. All time values are expressed in consistent time units.

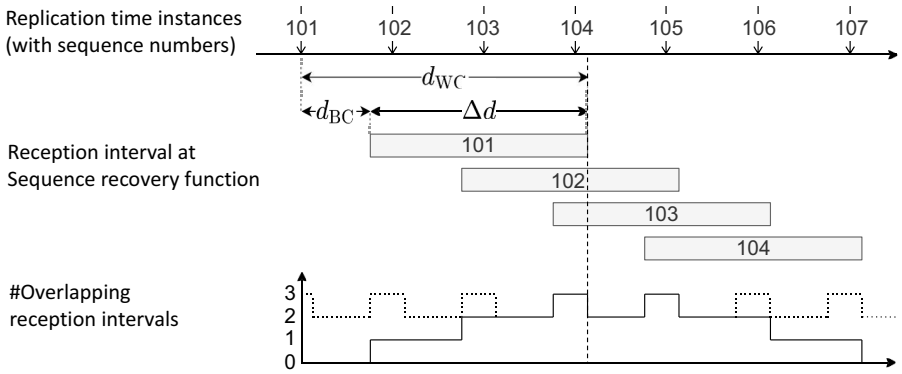


Figure X-2—Reception window illustration for Periodic Stream.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

X.3 Consequences of Parameter Settings

The following items describe the impact of parameters in the context of the Sequence recovery function. Parameters in the context of the Individual recovery function are not addressed.

a) Selecting MatchRecoveryAlgorithm (7.4.3.5) via the frerSeqRcvyAlgorithm parameter is only appropriate for Intermittent Streams (item c in 7.1.1). If applied to other types of Streams, this algorithm can incorrectly forward replicates. Therefore, it is important to identify whether a given Stream qualifies as intermittent.
b) If the frerSeqRcvyHistoryLength parameter is configured too small, the first received packet of a given Sequence Number can be classified as rogue and get discarded, since not all Member Streams in transit are considered.
c) If the frerSeqRcvyResetMSec parameter is configured with a value that is too low, this can result in unnessary resets and replicates can be forwarded due to the premature reset of the SequenceHistory (7.4.3.2.2). If the value is too high, and the reset is not triggered in time, the first received packet of a given Sequence Number after returning to normal operation can be classified as rogue and get discarded.

X.4 Parameter Configuration

This section provides configuration guidelines for parameters of the Sequence recovery function. Each parameter is discussed for Periodic and Interval-constrained Streams, followed by adaptions for Streams with more than one packet per interval.

X.4.1 Recovery Algorithm Selection

For the elimination of replicates, two recovery algorithms are available: the MatchRecoveryAlgorithm and the VectorRecoveryAlgorithm. The selection is done with the frerSeqRcvyAlgorithm parameter.

The MatchRecoveryAlgorithm stores a single Sequence Number and eliminates only replicates with that exact value. All other packets are accepted and forwarded. This method is applicable to Intermittent Streams; for all other Stream types, it can result in forwarding replicates. If the following equations are true, a Stream can be classified as Intermittent.

X.4.1.1 Periodic Streams

For Periodic Streams, a sufficient condition for Intermittent Stream behavior is that the transmission interval T_S exceeds the reception window Δd, including the maximum deviation V_S between the current and the next packet:

$$T_S > \Delta d + V_S \quad \text{(X-1)}$$

This ensures that all replicates of a Sequence Number have arrived before a new Sequence Number is received, as the reception intervals do not overlap.

X.4.1.2 Interval-constrained Streams and Multiple Packets

For Streams with multiple packets per interval (i.e., N_S>1) and for Interval-constrained Streams, the MatchRecoveryAlgorithm is inapplicable, since overlapping reception windows are possible. The VectorRecoveryAlgorithm should be used instead.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

X.4.2 Sequence History Length

If a Stream requires the VectorRecoveryAlgorithm, the frerSeqRcvyHistoryLength parameter must be configured. This parameter represents the number of Sequence Numbers that are retained by the sequence window of the Sequence recovery function. The sequence window is defined as the interval:

[RecovSeqNum - frerSeqRcvyHistoryLength + 1, (X-2)
RecovSeqNum + frerSeqRcvyHistoryLength]

The parameter RecovSeqNum (7.4.3.2.3) is initialized with the first Sequence Number received.

Figure X-3 provides an example of a sequence window with frerSeqRcvyHistoryLength = 5 and illustrates the behavior of the VectorRecoveryAlgorithm upon reception of the next packet. Packets with Sequence Numbers within the sequence window and greater than RecovSeqNum cause an update of RecovSeqNum. Packets that fall within the sequence window are handled correctly; packets that fall outside the window are marked as rogue and discarded, even if they are the first received packet of a given Sequence Number.

In the following, L denotes the value of frerSeqRcvyHistoryLength. This section discusses the behavior of the VectorRecoveryAlgorithm under normal operation. The behavior after resets will be discussed in Section X.4.3.

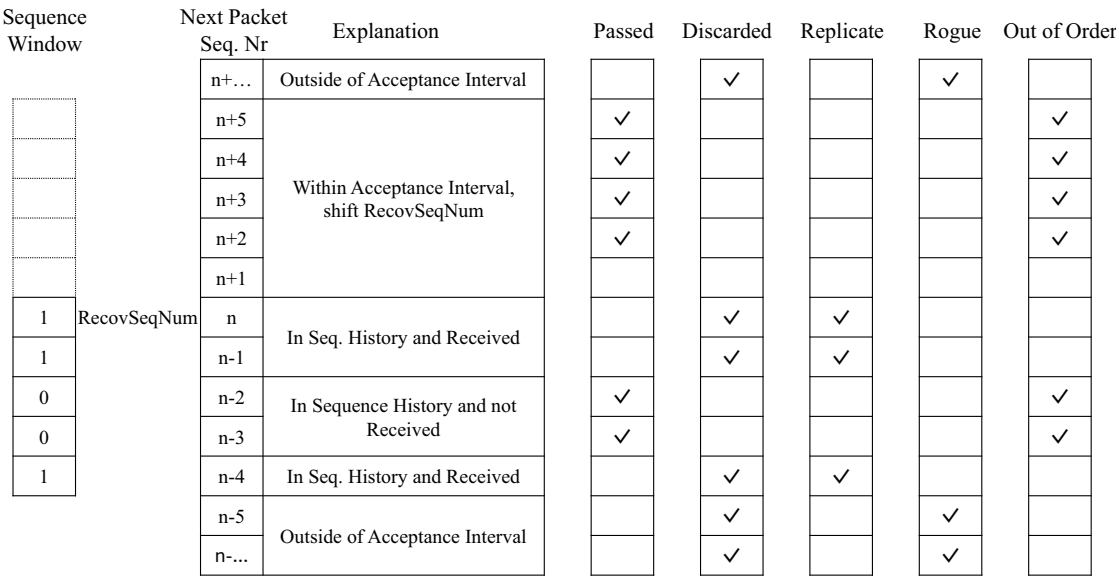


Figure X-3—Illustration of the VectorRecoveryAlgorithm. Headers indicate behavior, not variable or parameter names.

X.4.2.1 Periodic Streams

The following derivation applies to Periodic Streams with $N_S = 1$.

a) The sequence window length should contain all Sequence Numbers that could be received concurrently. This is defined by overlapping reception intervals, as illustrated with the function at the bottom of Figure X-2. Only the reception of new packets triggers updates in the sequence window.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

b) With packet transmissions occurring every T_S time units, the number of overlapping reception intervals and thus the required sequence window length is given by:

$$L \geq \left\lfloor \frac{\Delta d}{T} \right\rfloor + 1 \quad \text{(X-3)}$$

where ⌊x⌋ denotes the floor function, meaning the greatest integer less than or equal to x. **Example**. In the example illustrated in Figure X-2., N = 3, and the current RecovSeqNum is 101. A packet with Sequence Number 104 can be accepted before Sequence Numbers 102 or 103 have arrived. At the same time, Sequence Number 99 can still arrive after Sequence Number 101 has been accepted.

c) Including the maximum timing deviation V_S results in:

$$L \geq \left\lfloor \frac{(\Delta d + V_S)}{T_S} \right\rfloor + 1 \quad \text{(X-4)}$$

Lower values should not be used. Higher values increase memory overhead.

X.4.2.2 Interval-constrained Streams

In this case, packets can occur at any time during the interval T_S, leading to:

$$L \geq \left\lfloor \frac{(\Delta d + V_S)}{T_S} \right\rfloor + 2 \quad \text{(X-5)}$$

X.4.2.3 Multiple Packets

For Streams with N_S>1, each overlapping Sequence Number can correspond to multiple packets. The worst-case scenario occurs when one packet is sent at the beginning of an interval and the remaining N_S - 1 packets at the end. The required sequence window length becomes:

$$L' \geq N_S \times L \quad \text{(X-6)}$$

X.4.3 Reset Timer Configuration

The Sequence recovery function is reinitialized with the SequenceRecoveryReset (7.4.3.3) function if no packet is accepted and forwarded for the duration defined by frerSeqRcvyResetMSec. This means that after the reset is triggered, the history is set to all zeros and RecovSeqNum will be set to the next packet received. This timeout prevents packets from being incorrectly handled after interruptions, due to, e.g., misbehaving or failing nodes. Each time a new packet is accepted, the timer is restarted.

If the timeout is configured too short, the sequence window can reset prematurely and allow replicates to be forwarded. If the timeout is too long, the first received packet of a given Sequence Number after an interruption can be discarded as being outside the current sequence window. Thus, the reset should be triggered as early as possible, but not before all possible replicates have been received.

This section does not address the maximum duration of potential interruptions. Instead, it specifies the length of the reset timer, defined by the frerSeqRcvyResetMSec parameter and denoted as R in the following.

X.4.3.1 Periodic Streams

The following derivation applies to Periodic Streams with N_S=1:

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

a) To prevent the forwarding of replicates, a safe reset can occur when no more replicates of the current Sequence Number can arrive. Assuming the first packet of a given Sequence Number arrives at the beginning of Δd, no replicates will be forwarded if the timeout is:

$$R > \Delta d + V_S \quad \text{(X-7)}$$

b) However, if Δd is small and no overlap exists between reception interval of Sequence Numbers, this can lead to unnecessary resets after each packet. To avoid this, the timer can wait until the entire interval of the next packet has passed:

$$R = \Delta d + T_S + V_S \quad \text{(X-8)}$$

This value ensures that the Sequence recovery function resets only when no further replicates are expected, without delaying the acceptance of new Sequence Numbers. Higher values are not recommended, as they can cause the first received packet of a given Sequence Number after interruptionsto be discarded.

X.4.3.2 Interval-constrained Streams

In this case, packets can occur at any time during the interval T_S, leading to:

$$R = \Delta d + 2 \times T_S + V_S \quad \text{(X-9)}$$

X.4.3.3 Multiple Packets

The reset timer only depends on the timing of the packet arrivals, not on the number of packets. Therefore, the equations above apply for all values of N_S.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.

Annex Y

(informative)

Bibliography

<< Insert new reference, here numbered as [B1], renumbering, and updating cross-references as necessary. >>

[B1] L. Maile, D. Voitlein, K.-S. Hielscher, and R. German, “Ensuring reliable and predictable behavior of IEEE 802.1CB frame replication and elimination,” in Proc. IEEE International Conference on Communications, Seoul, Korea, Republic of, 2022, pp. 2706-2712, doi: 10.1109/ ICC45855.2022.9838905.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards Draft, subject to change.